DERIVATION OF THE WEAK FORM OF THE DIFFUSION EQUATION AND FEM

This poster derives the weak formulation of the one-dimensional diffusion equation and applies the Galerkin Finite Element Method (FEM) to approximate transient heat conduction. The poster discretizes the resulting weak formulation using linear basis functions and constructs the mass and stiffness matrices. The Forward and Backward Euler methods are introduced to discretize the time term. The results of the Forward Euler Scheme are compared with the analytical solution obtained using the Fourier sine series in MATLAB. This study shows how the weak formulation and the Finite Element Method can be used to approximate solutions to transient heat conduction problems, particularly the problems involving complex geometries, discontinuities, or solutions where second-order derivatives may not exist.